

BeeSkilled – Cloud Computing with AWS

Week 1 – Submission Document

Assignment 1 + Assignment 2 + Mini Project

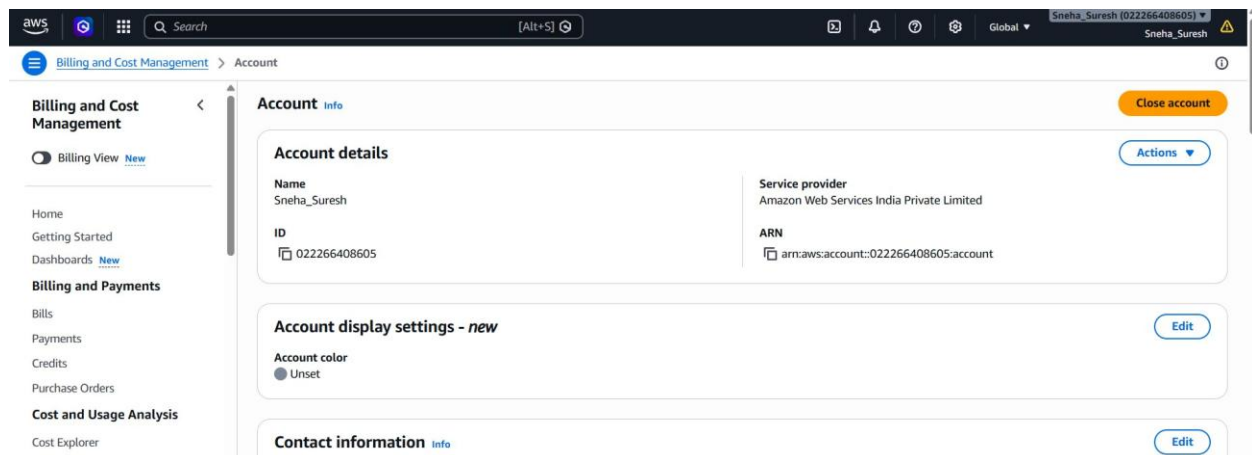
Name: Sneha Suresh

PART 1 – Assignment 1: AWS Account Setup & IAM Configuration

This section provides evidence of completing Assignment 1 — creating and configuring an AWS account, setting up IAM users and groups, and exploring the AWS Management Console.

Step 1 – AWS Account Details

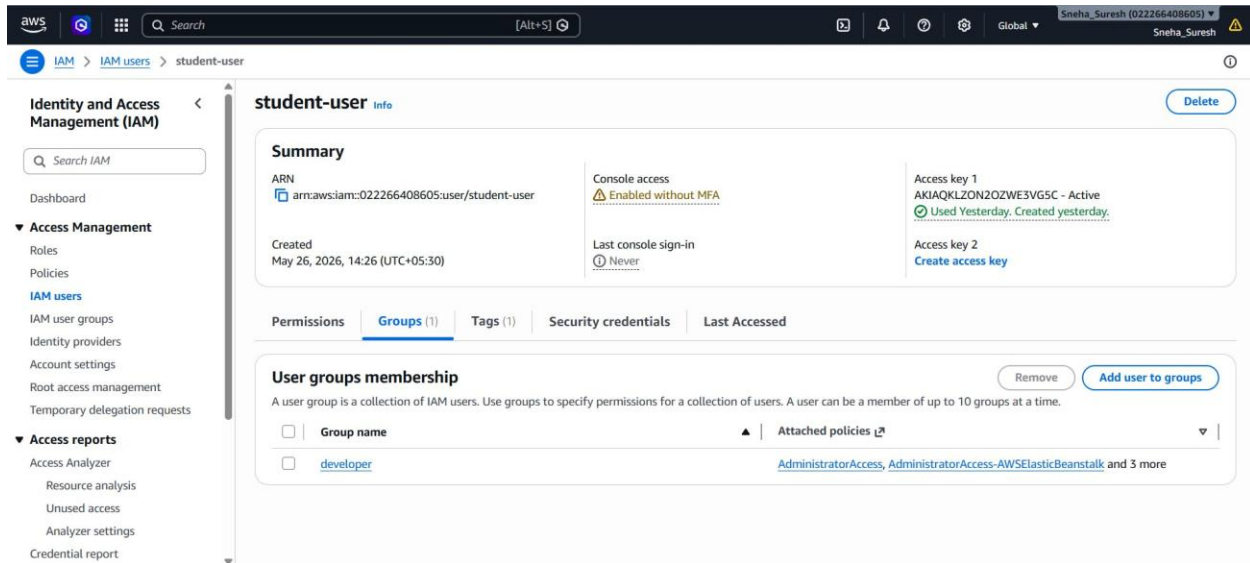
An AWS Free Tier account was set up under the name Sneha_Suresh (Account ID: 022266408605). The account is registered with Amazon Web Services India Private Limited.



Screenshot 1 – AWS Billing & Account page showing account name and ID

Step 2 – IAM User Created: student-user

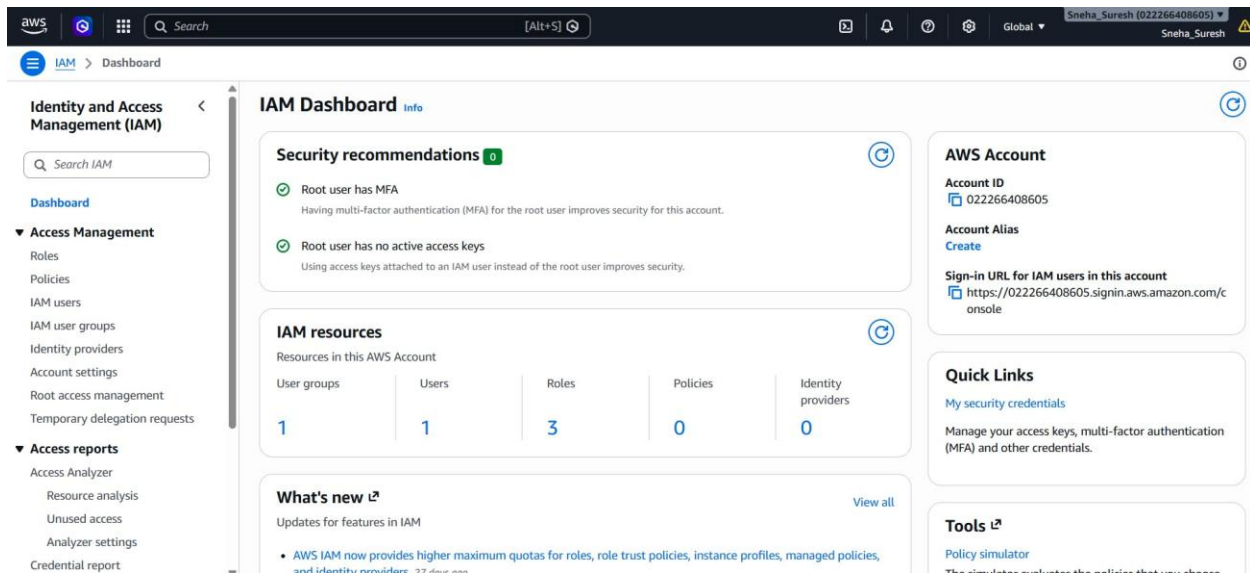
An IAM user named 'student-user' was created with console access enabled. The user was created on May 26, 2026, and has an active access key. The user is assigned to the 'developer' group with AdministratorAccess and 3 additional policies.



Screenshot 2 – IAM user 'student-user' with group membership and attached policies

Step 3 – IAM Dashboard Overview

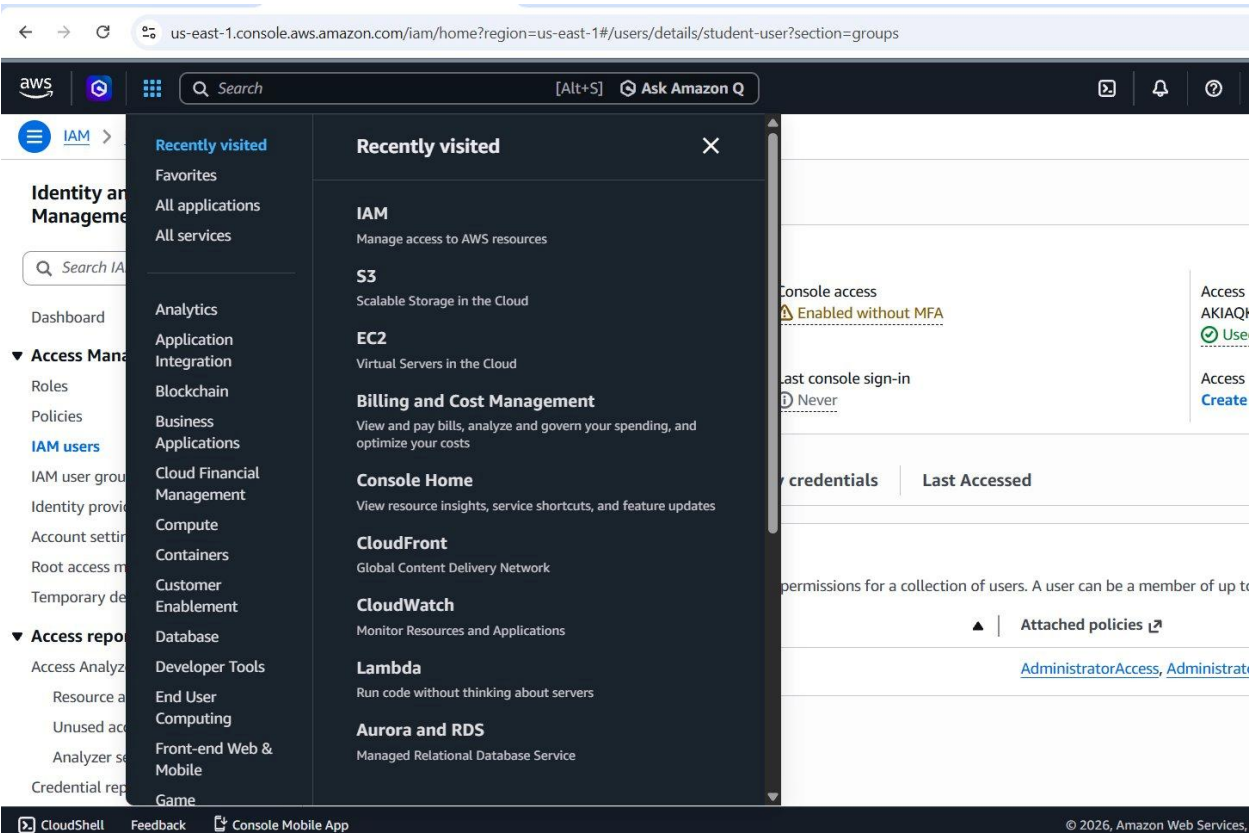
The IAM Dashboard confirms the account configuration: 1 User Group ('developer'), 1 User ('student-user'), and 3 Roles. Security recommendations show that Root user MFA is enabled and no active access keys exist on the root account — both are best practices.



Screenshot 3 – IAM Dashboard showing 1 user group, 1 user, 3 roles, and security recommendations

Step 4 – AWS Management Console Explored

The AWS Management Console was explored, including recently visited services: IAM, S3, EC2, Billing and Cost Management, CloudFront, CloudWatch, Lambda, and Aurora/RDS. This demonstrates familiarity with the console navigation and core services.



Screenshot 4 – AWS Management Console showing recently visited services

Assignment 1 – IAM Setup Summary

#	Step	Details
1	Open IAM Console	Navigated to IAM from the AWS Management Console search bar.
2	Create IAM User	Created a new user named 'student-user' with console and programmatic access enabled.
3	Assign Permissions	Attached AdministratorAccess and related policies to the user.
4	Create User Group	Created a group named 'developer', attached AdministratorAccess and 3 more policies, added student-user to the group.
5	Verify Configuration	Confirmed IAM Dashboard shows 1 User, 1 Group, 3 Roles. Root user MFA enabled.

PART 2 – Assignment 2: Core AWS Services Summary Report

Objective

The objective of this assignment is to identify and describe the key AWS services — EC2, S3, RDS, and Lambda — including their core functions, service categories, and real-world use cases.

Overview of Core AWS Services

Amazon Web Services (AWS) provides over 200 cloud services globally. The four foundational services explored in this report are summarised in the table below:

Service	Full Name	Category	Core Function
EC2	Elastic Compute Cloud	Compute (IaaS)	Scalable virtual servers. Choose OS, CPU, RAM. Used to host apps and run workloads.
S3	Simple Storage Service	Object Storage	Store and retrieve any amount of data. Used for files, backups, and static websites.
RDS	Relational Database Service	Database (PaaS)	Managed relational DB (MySQL, PostgreSQL, etc.). Automates backups, patching, scaling.
Lambda	AWS Lambda	Serverless Compute	Run code without servers. Triggered by events. Billed per execution and duration.

1. Amazon EC2 – Elastic Compute Cloud

Amazon EC2 is a core IaaS offering that provides scalable virtual machines (instances) in the cloud. Users have full control over the operating system, CPU, memory, and storage configuration.

Key features:

- Wide variety of instance types (e.g., t2.micro for Free Tier, t3.medium for production)
- Supports both Linux and Windows operating systems
- Pay-as-you-go and reserved pricing options
- Integrates with Auto Scaling and Elastic Load Balancing for high availability

2. Amazon S3 – Simple Storage Service

Amazon S3 is an object storage service for storing and retrieving any amount of data. It offers 99.999999999% (eleven nines) durability.

Key features:

- Bucket-based organisation with globally unique names
- Supports versioning, lifecycle policies, and access control lists
- Can host static websites directly from a bucket
- Fine-grained access control using bucket policies and IAM

3. Amazon RDS – Relational Database Service

Amazon RDS is a managed database service that simplifies deploying and operating relational databases.

Key features:

- Supports MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server
- Automated backups and point-in-time recovery
- Multi-AZ deployment for fault tolerance and high availability
- Read replicas to scale read-heavy database workloads

4. AWS Lambda

AWS Lambda is a serverless compute service that runs code in response to events without requiring server provisioning or management.

Key features:

- Supports Node.js, Python, Java, Go, Ruby, and more
- Event-driven triggers from S3, API Gateway, DynamoDB, SNS, etc.
- Scales from zero to thousands of concurrent executions automatically
- Billed only per request and compute duration (in milliseconds)

Conclusion – Assignment 2

EC2, S3, RDS, and Lambda are the four pillars of the AWS ecosystem covering compute, storage, database, and serverless workloads. Understanding these services is essential for architecting scalable and cost-efficient cloud solutions.

PART 3 – Mini Project: IAM & S3 Hands-On

Project Objective

The goal of this mini project is to gain hands-on experience with AWS IAM and Amazon S3 by creating an IAM user with specific permissions, setting up an S3 bucket, uploading static files, and configuring public access through a bucket policy.

Tools & Services Used

- AWS Management Console
- AWS IAM (Identity and Access Management)
- Amazon S3 (Simple Storage Service)

Part A – IAM User Setup (Completed)

As demonstrated in Assignment 1 above, the IAM user 'student-user' was created and added to the 'developer' group with appropriate permissions. Screenshots are provided in Part 1.

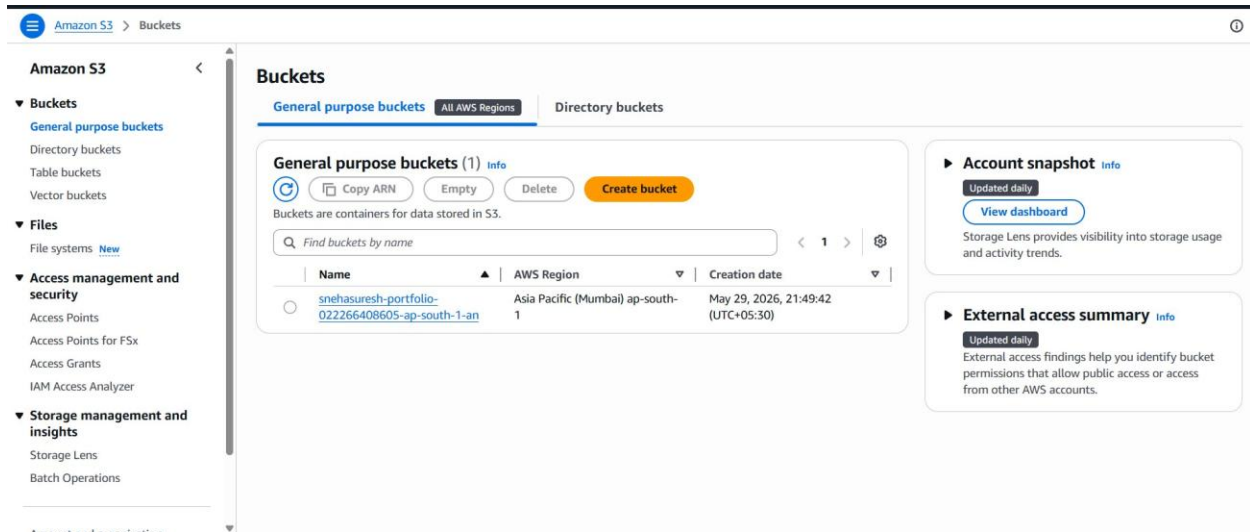
Part B – S3 Bucket Setup & Static File Hosting

The following steps were performed to create an S3 bucket, upload files, and configure public access:

#	Step	Details
1	Open S3 Console	Navigated to S3 from the AWS Management Console.
2	Create S3 Bucket	Created a bucket with a unique name. Selected ap-south-1 (Mumbai) as the region.
3	Upload Static Files	Uploaded index.html and a sample image file to the bucket.
4	Configure Public Access	Disabled 'Block all public access' setting under bucket permissions.
5	Apply Bucket Policy	Added a bucket policy to allow s3:GetObject for all principals (*) on all objects.
6	Test Public Access	Copied the object URL and opened it in the browser — static file loaded successfully.

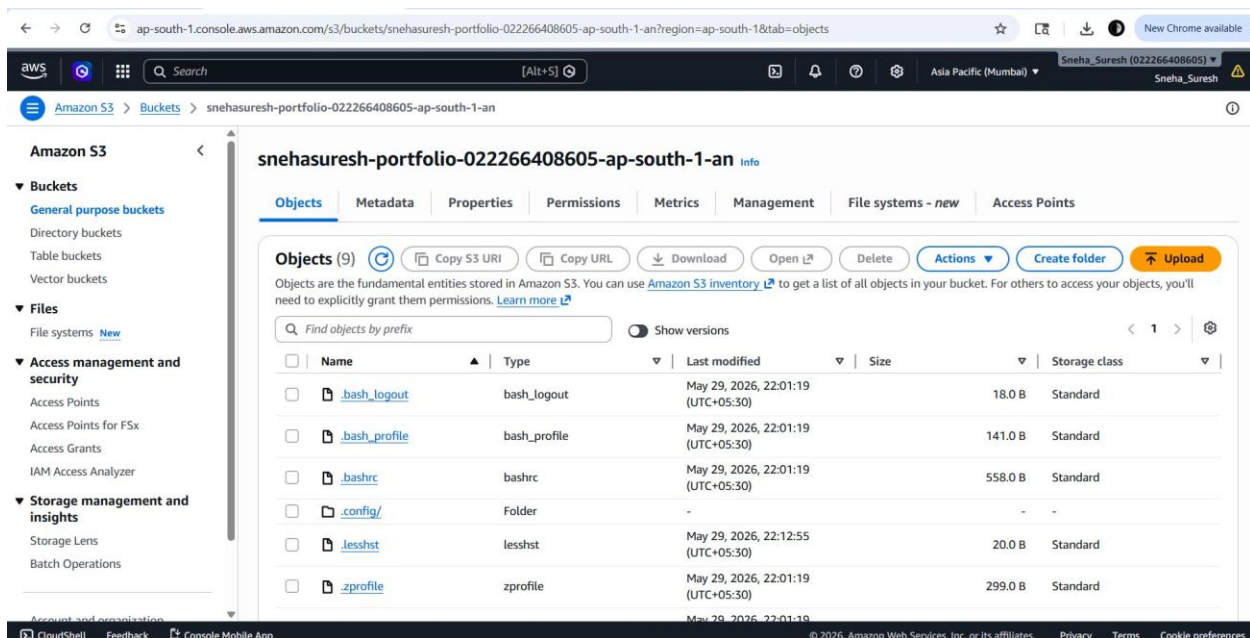
S3 Bucket Screenshots

The S3 bucket named 'snehasuresh-portfolio-022266408605-ap-south-1-an' was successfully created in the Asia Pacific (Mumbai) ap-south-1 region on May 29, 2026.



Screenshot 5 – S3 Buckets list showing the created bucket in ap-south-1 (Mumbai) region

9 objects were successfully uploaded to the bucket including static files (.bash_logout, .bash_profile, .bashrc, .lessht, .zprofile) and a configuration folder (.config/).



Screenshot 6 – S3 bucket contents showing 9 uploaded objects

S3 Bucket Policy Applied

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:GetObject",
```

```
"Resource": "arn:aws:s3:::your-bucket-name/*"  
  }]  
}
```

Outcome & Observations

- Successfully created IAM user 'student-user' with scoped permissions following least privilege principle
- IAM Dashboard confirmed: 1 User, 1 Group, 3 Roles — all correctly configured
- S3 bucket configured to serve static files publicly via bucket policy
- Static HTML file was accessible via the S3 object URL in the browser
- Learned the difference between ACL-based and policy-based access control in S3

Learning Outcome

Through this mini project and assignments, a solid understanding was gained of AWS IAM for access management and Amazon S3 for object storage and static website hosting. The hands-on experience reinforced cloud computing fundamentals including security best practices, bucket policy configuration, IAM user/group management, and the AWS global infrastructure setup.